Confidence Intervals (Small Sample)

Lecture 11

Small-sample Confidence Intervals for a Population Mean

- When the sample size is large, by C.L.T.

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

- When the sample size is small, there are no good general methods for finding confidence intervals

- However, when the population is approximately normal, a probability distribution called *Student's t distribution* can be used to compute confidence intervals for a population mean

Student's t Distribution

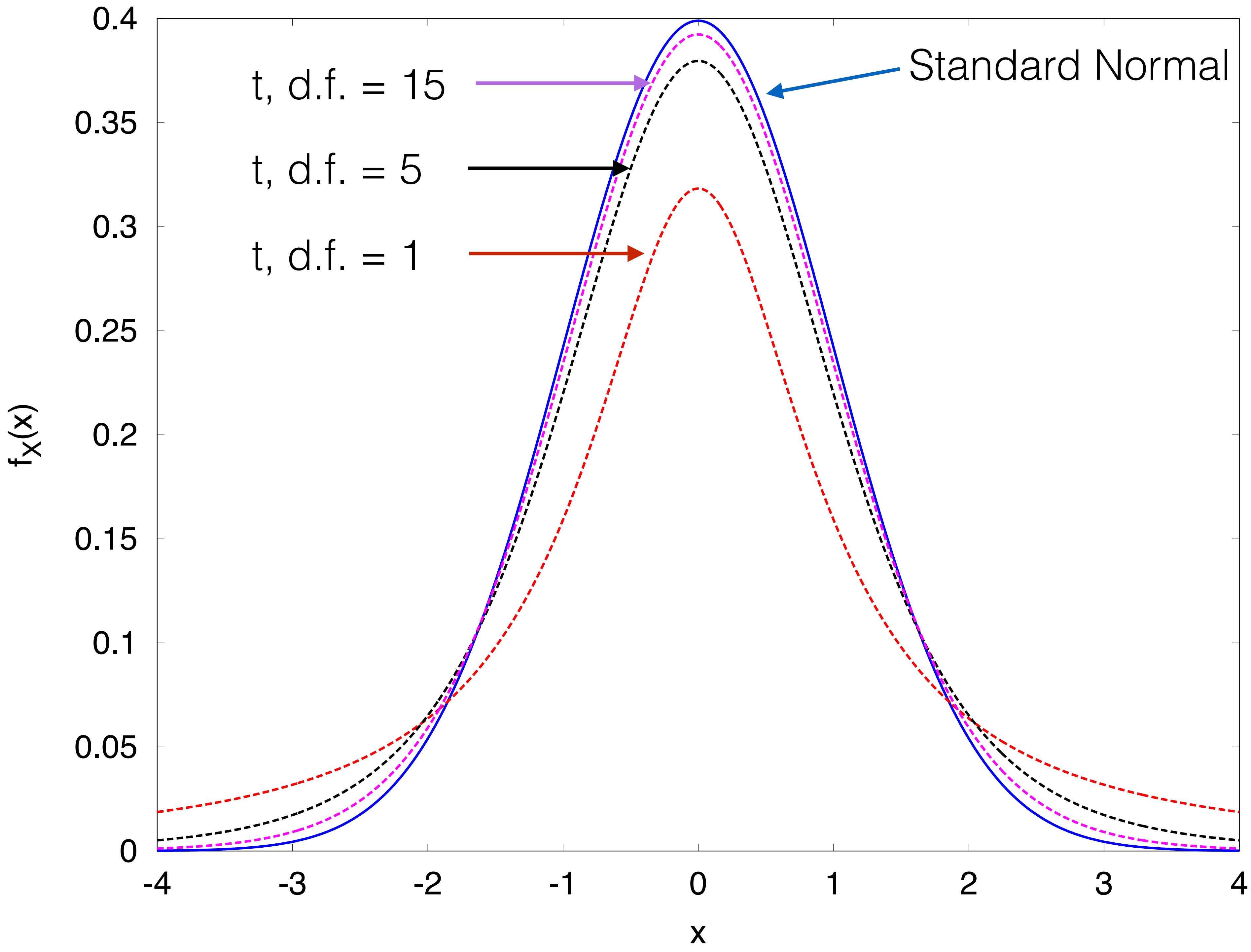
- Let $X_1, X_2, \ldots, X_n$ be a small ($n < 30$) random sample from a *normal population* with mean μ. Then the quantity

$$t = \frac{\overline{X} - \mu}{s/\sqrt{n}}$$

has a *Student's t distribution with n-1 degree of freedom*, denoted by t_{n-1}

- When n is large, the distribution of the quantity $\frac{\overline{X} - \mu}{s/\sqrt{n}}$ is very close to normal, so the normal curve can be used rather than the Student's t

Student's t Distribution



Student's t Table

source: http://www.stat.ufl.edu/~athienit/Tables/tables.pdf

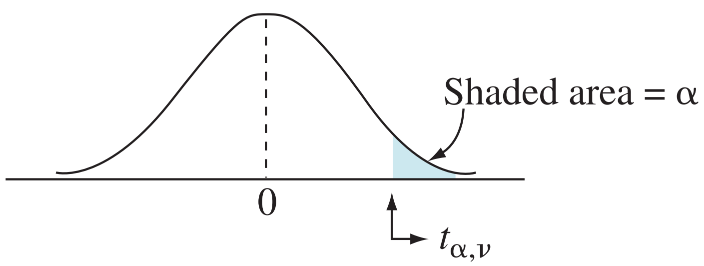


TABLE 2

Percentage points of Student's t distribution

df/α =	.40	.25	.10	.05	.025	.01	.005	.001	.0005
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657	318.309	636.619
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.258	0.692	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.257	0.686	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.256	0.686	1.321	1.717	2.074	2.508	2.819	3.505	3.792

Example 1

- A random sample of size 12 is to be drawn from a normal population with mean μ. The Student's t statistic is to be computed. What is the probability that $t > 1.796$?

C.I. Using the Student's t Distribution

- Let X_1, X_2, …, X_n be a small (n < 30) random sample from a *normal population* with mean μ. Then a level 100(1-α)% confidence interval for μ is

$$\overline{X} \pm t_{n-1,\alpha/2} \frac{s}{\sqrt{n}}$$

Example 2

- A sample set of data from a normal population is as follows. Find the 95% confidence interval for the population mean.

19	21	18	19	20
12	25	21	21	12
20	12	15	17	23

Example 2

- A sample set of data from a normal population is as follows. Find the 99% confidence interval for the population mean.

19	21	18	19	20
12	25	21	21	12
20	12	15	17	23

Student's t One-sided C.I.

- Let X_1, X_2, …, X_n be a small ($n < 30$) random sample from a *normal population* with mean μ. Then a level 100(1- α)% lower confidence bound for μ is

$$\overline{X} - t_{n-1,\alpha} \frac{s}{\sqrt{n}}$$

and level 100(1- α)% upper confidence bound or μ is

$$\overline{X} + t_{n-1,\alpha} \frac{s}{\sqrt{n}}$$

Example 3

- A sample of 10 snack boxes is drawn from a population. The sample mean of the weight is 254 grams, and the sample standard deviation is 11 grams. If the population weight has a normal distribution. Find the one-sided 90% lower confidence bound for the mean of the weight.

If σ is known

- Use Z, not t, if σ is known

- Let X_1, X_2, …, X_n be a random sample (of any size) from a *normal population* with mean μ. If the standard deviation σ is known, then a level 100(1-α)% confidence interval for μ is

$$\overline{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$